

## EXPERIMENT 2

### Area of a circle and stages of c compilation

We will create and run a C program in the Kali Linux terminal using the cc compiler. The program calculates the area of a circle and demonstrates the various stages involved in compiling a C program.

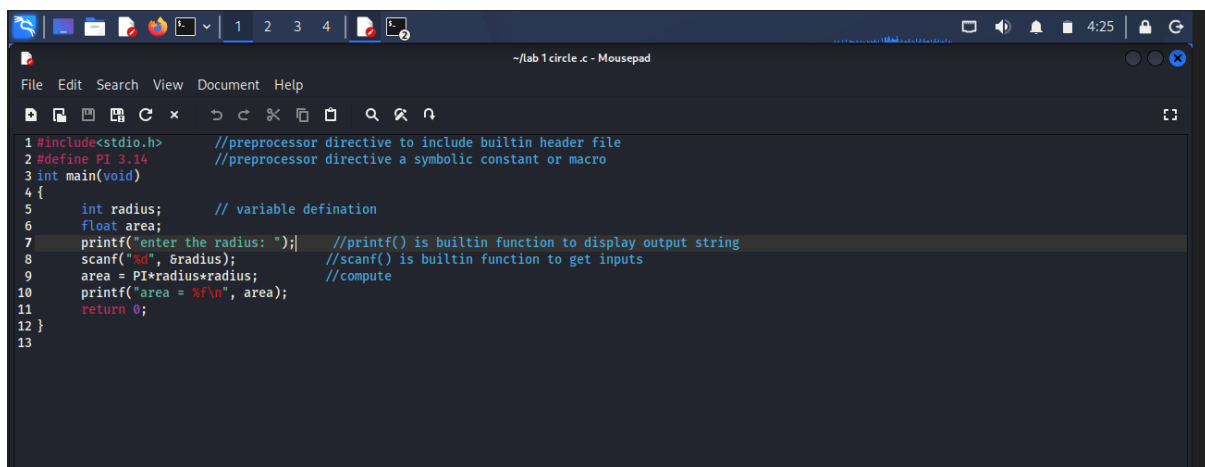
Formula for the area of a circle:

$$\text{Area} = \pi r^2$$

Here, r represents the radius of the circle, and  $\pi$  (PI) is approximately equal to 3.14.

#### 1. Create the C program :

Open the Kali Linux terminal and create a C source file using the command `vi circle.c`. The file will open in the normal mode of the Vi editor. Press `i` to enter insert mode and type the C program. After completing the program, press the `Esc` key to return to normal mode. Finally, type `:wq` and press `Enter` to save the file, close the editor, and return to the terminal.



```
1#include<stdio.h>           //preprocessor directive to include builtin header file
2#define PI 3.14             //preprocessor directive a symbolic constant or macro
3int main(void)
4{
5    int radius;              // variable defination
6    float area;
7    printf("enter the radius: "); //printf() is builtin function to display output string
8    scanf("%d", &radius);      //scanf() is builtin function to get inputs
9    area = PI*radius*radius;    //compute
10   printf("area = %f\n", area);
11   return 0;
12}
13
```

## 2. Compile and Save the Intermediate Files

Compile the C program using the following command:

```
cc -save-temps circle.c
```

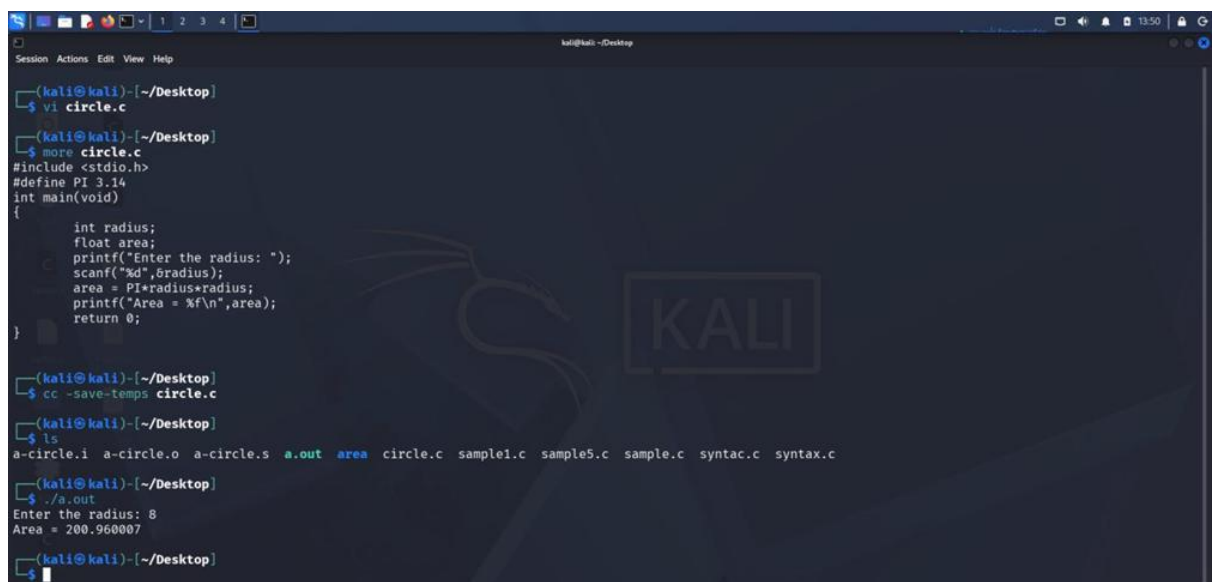
The `-save-temps` option instructs the compiler to retain the intermediate files generated during the different compilation stages.

To view these files, use:

```
ls
```

The following files will generally be displayed:

- `circle.c` — source file
- `circle.i` — preprocessed file
- `circle.s` — assembly file
- `circle.o` — object file
- `a.out` — final executable file



```
kali@kali: ~/Desktop
$ vi circle.c
$ more circle.c
#include <stdio.h>
#define PI 3.14
int main(void)
{
    int radius;
    float area;
    printf("Enter the radius: ");
    scanf("%d",&radius);
    area = PI*radius*radius;
    printf("Area = %f\n",area);
    return 0;
}
$ cc -save-temps circle.c
$ ls
a-circle.i a-circle.o a-circle.s a.out area circle.c sample1.c sample5.c sample.c syntac.c syntax.c
$ ./a.out
Enter the radius: 8
Area = 200.960007
$
```

### 3. Stages of Compilation

#### Stage 1 — Preprocessing:

The preprocessor handles directives beginning with #, such as:

```
#include <stdio.h>
```

It expands the required header files and macros and generates the preprocessed file circle.i.

#### Stage 2 — Compilation:

The compiler translates the preprocessed C code into assembly-language instructions. The output of this stage is stored in circle.s.

#### Stage 3 — Assembly:

The assembler converts the assembly instructions into relocatable machine code and produces the object file circle.o.

#### Stage 4 — Linking:

The linker combines circle.o with the required system libraries. It resolves references to library functions such as printf() and scanf() and creates the final executable file a.out.

### 4. Execution of the Program

Run the executable using:

```
./a.out
```

Example output:

Enter the radius of circle: 8

Area of circle = 200.960007

## Result:

The C program was successfully created, compiled using the `cc` compiler, and executed in the Kali Linux terminal. The stages of preprocessing, compilation, assembly, and linking were examined through the intermediate files retained using the `-save-temps` option.